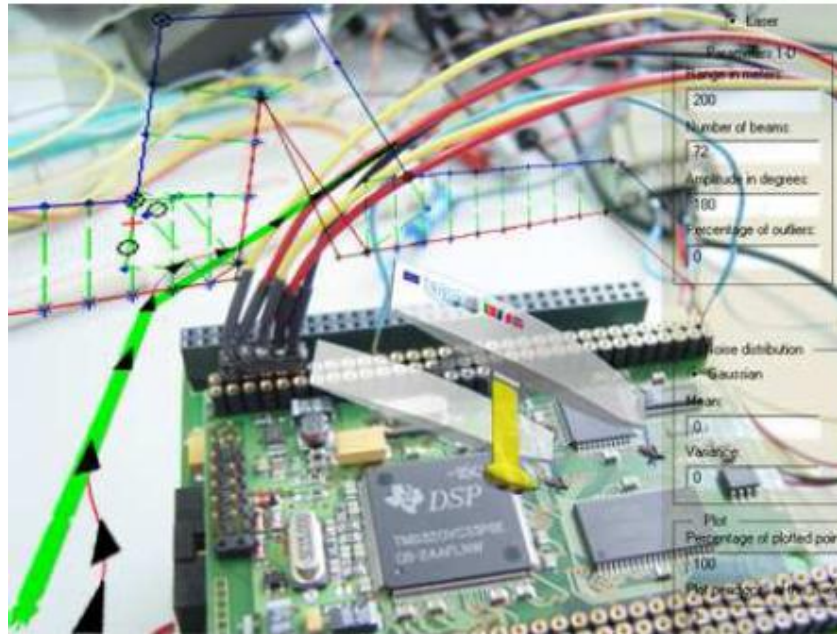
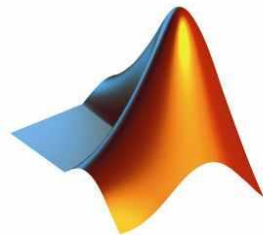




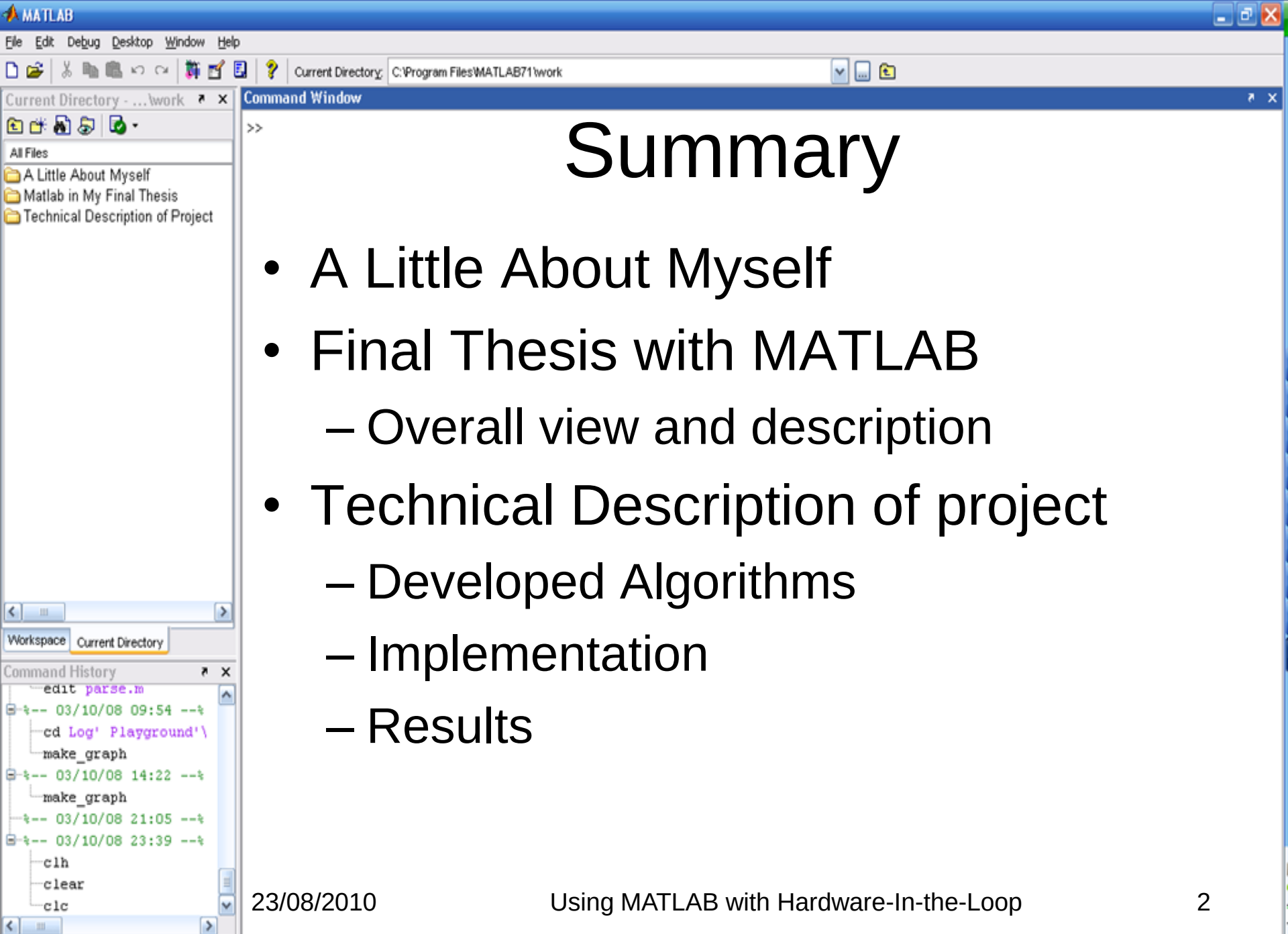
Using MATLAB with Hardware-In-the-Loop

“It is always wise to look ahead, but
difficult to look further than you can see.”

Winston Churchill



The MathWorks



The image shows a MATLAB software interface. The top menu bar includes File, Edit, Debug, Desktop, Window, and Help. Below the menu is a toolbar with icons for file operations and a question mark. The 'Current Directory' is set to 'C:\Program Files\MATLAB71\work'. On the left, the 'Current Directory' pane shows a file tree with folders: 'A Little About Myself', 'Matlab in My Final Thesis', and 'Technical Description of Project'. Below this is the 'Command History' pane, which lists several commands: 'edit parse.m', 'cd Log\ Playground\'', 'make_graph', and 'clear'. The main 'Command Window' on the right displays a large 'Summary' title and a bulleted list of topics. At the bottom of the MATLAB window, the Windows 'Start' button is visible.

Summary

- A Little About Myself
- Final Thesis with MATLAB
 - Overall view and description
- Technical Description of project
 - Developed Algorithms
 - Implementation
 - Results

23/08/2010 Using MATLAB with Hardware-In-the-Loop 2

MATLAB

File Edit View Debug Desktop Window Help

Current Directory: C:\Program Files\MATLAB71\work\A Little About Myself

Current Directory - ...Myself

All Files

Education

Altran CIS

Oceanscan

Command Window

A Little about myself

- 5 Year Degree in Electrical Engineering and Computer Science
 - Control Theory and Image Processing
 - Operating Systems
- Average 14.45 out 20 – top 10%

Workspace Current Directory

Command History

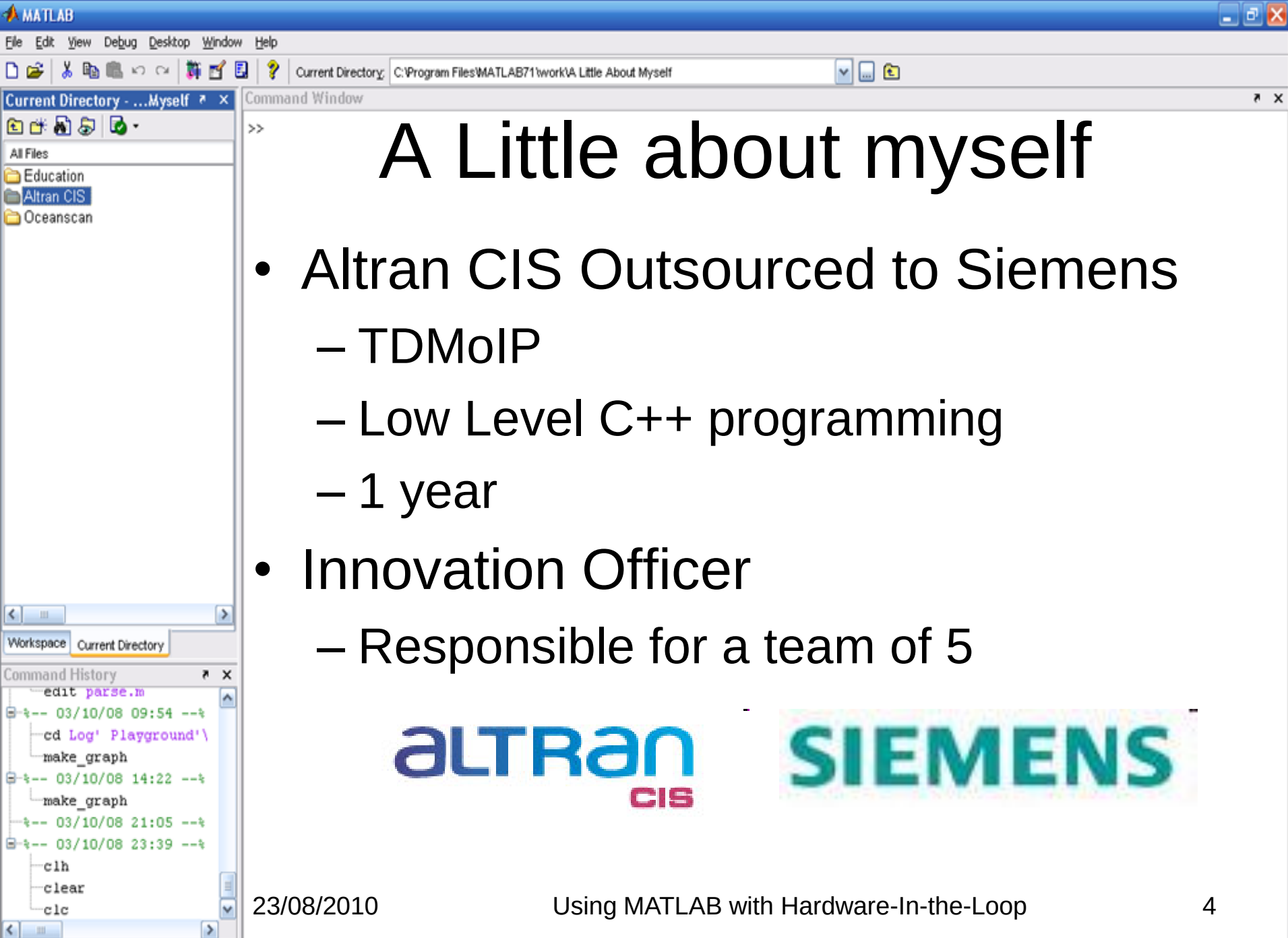
```
edit parse.m
-- 03/10/08 09:54 --
cd Log\ Playground\
make_graph
-- 03/10/08 14:22 --
make_graph
-- 03/10/08 21:05 --
-- 03/10/08 23:39 --
cli
clear
clc
```



23/08/2010

Using MATLAB with Hardware-In-the-Loop

3



The image shows a MATLAB software window. The title bar says 'MATLAB'. The menu bar includes 'File', 'Edit', 'View', 'Debug', 'Desktop', 'Window', and 'Help'. The toolbar contains various icons for file operations and MATLAB-specific functions. The 'Current Directory' path is 'C:\Program Files\MATLAB71\work\A Little About Myself'. The 'Command Window' displays the title 'A Little about myself' and a bulleted list. The left sidebar shows a file explorer with folders 'Education', 'Altran CIS', and 'Oceanscan'. The 'Command History' pane at the bottom left shows a list of commands and their execution times.

A Little about myself

- Altran CIS Outsourced to Siemens
 - TDMoIP
 - Low Level C++ programming
 - 1 year
- Innovation Officer
 - Responsible for a team of 5



23/08/2010 Using MATLAB with Hardware-In-the-Loop

MATLAB

File Edit View Debug Desktop Window Help

Current Directory: C:\Program Files\MATLAB71\work\A Little About Myself

Current Directory - ...Myself

All Files

- Education
- Altran CIS
- Oceanscan

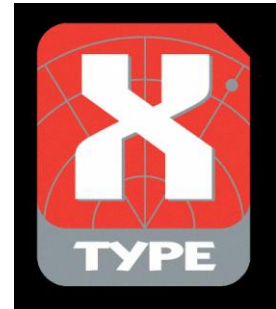
Workspace Current Directory

Command History

- edit parse.m
- 03/10/08 09:54 --> cd Log\ Playground\
- make_graph
- 03/10/08 14:22 --> make_graph
- 03/10/08 21:05 -->
- 03/10/08 23:39 --> clh
- clear
- clc

A Little about myself

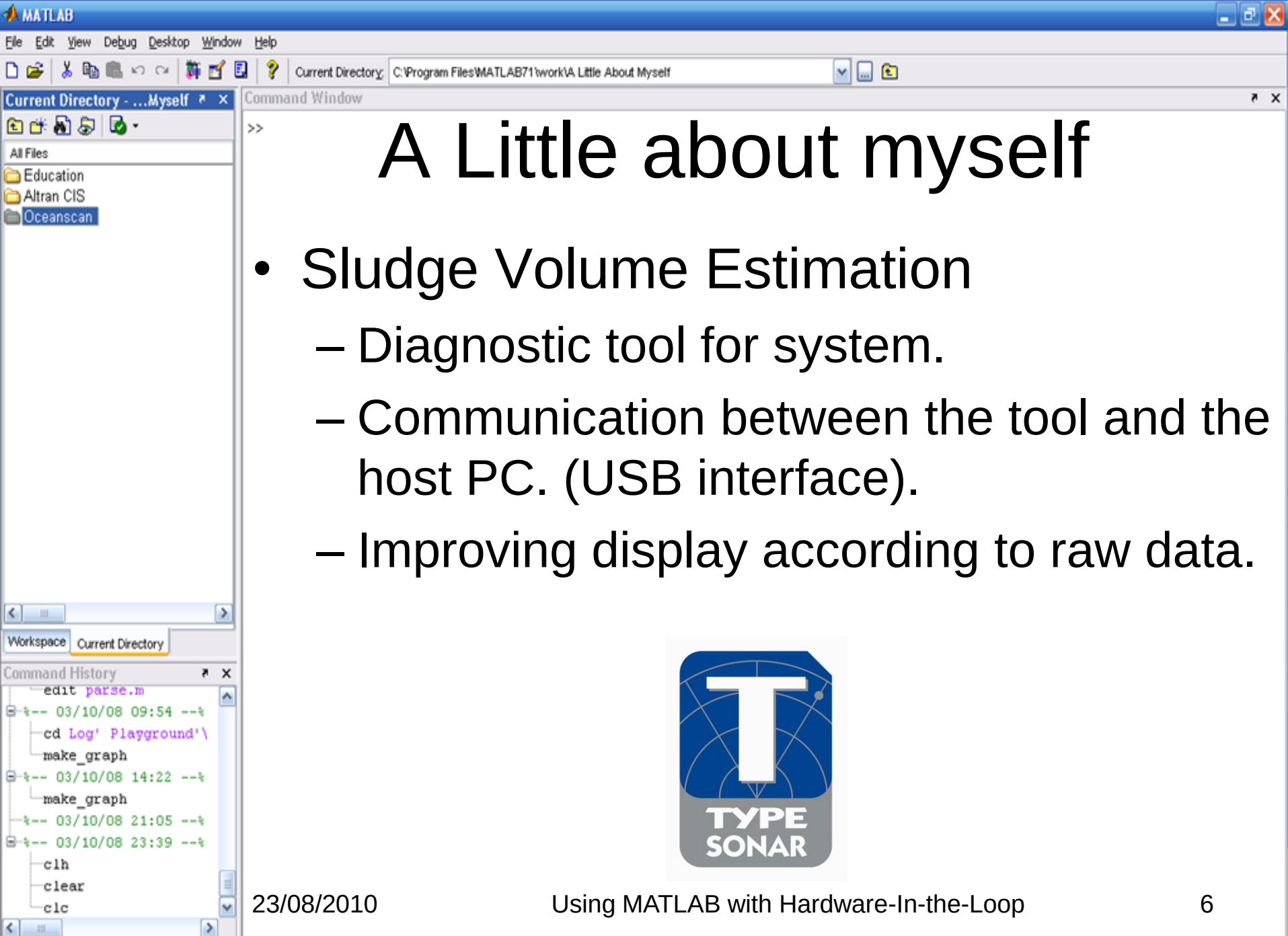
- Diver Detection System
 - Linux based real time data acquisition system.
 - Data acquisition system integration with GUI.
 - Analogue Outputs and Inputs - signal quality.
 - System Tester.
 - Data transfer from subsea head to host PC.
 - Diagnostic tool for system debug and performance benchmarking
 - User Interface with DirectX in C++ based on MFC.
 - Simulations in LabView.
 - On trials across the world.



23/08/2010

Using MATLAB with Hardware-In-the-Loop

5



The image shows a MATLAB software interface. The top menu bar includes File, Edit, View, Debug, Desktop, Window, and Help. The current directory is set to C:\Program Files\MATLAB71\work\A Little About Myself. The Command Window displays the title 'A Little about myself' and a bulleted list. The left sidebar shows a file explorer with folders Education, Altran CIS, and Oceanscan. The Command History window at the bottom lists recent commands and their execution times.

A Little about myself

- Sludge Volume Estimation
 - Diagnostic tool for system.
 - Communication between the tool and the host PC. (USB interface).
 - Improving display according to raw data.



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MATLAB

File Edit View Debug Desktop Window Help

Current Directory: C:\Program Files\MATLAB71\work\A Little About Myself

Current Directory - ...Myself

All Files

- Education
- Altran CIS
- Oceanscan

Workspace Current Directory

Command History

```
edit parse.m  
-- 03/10/08 09:54 --  
cd Log\ Playground\  
make_graph  
-- 03/10/08 14:22 --  
make_graph  
-- 03/10/08 21:05 --  
-- 03/10/08 23:39 --  
clh  
clear  
clc
```

A Little about myself

- General responsibilities
 - Software Version Control
 - Documentation, Specification and Template design



23/08/2010

Using MATLAB with Hardware-In-the-Loop

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MATLAB

File Edit View Debug Desktop Window Help

Current Directory: C:\Program Files\MATLAB71\work

Current Directory - ...work

All Files

- A Little About Myself
- Matlab in My Final Thesis
- Technical Description of Project

Workspace Current Directory

Command History

- edit parse.m
- 03/10/08 09:54 --> cd Log\ Playground\'
- make_graph
- 03/10/08 14:22 --> make_graph
- 03/10/08 21:05 -->
- 03/10/08 23:39 --> clh
- clear
- clc

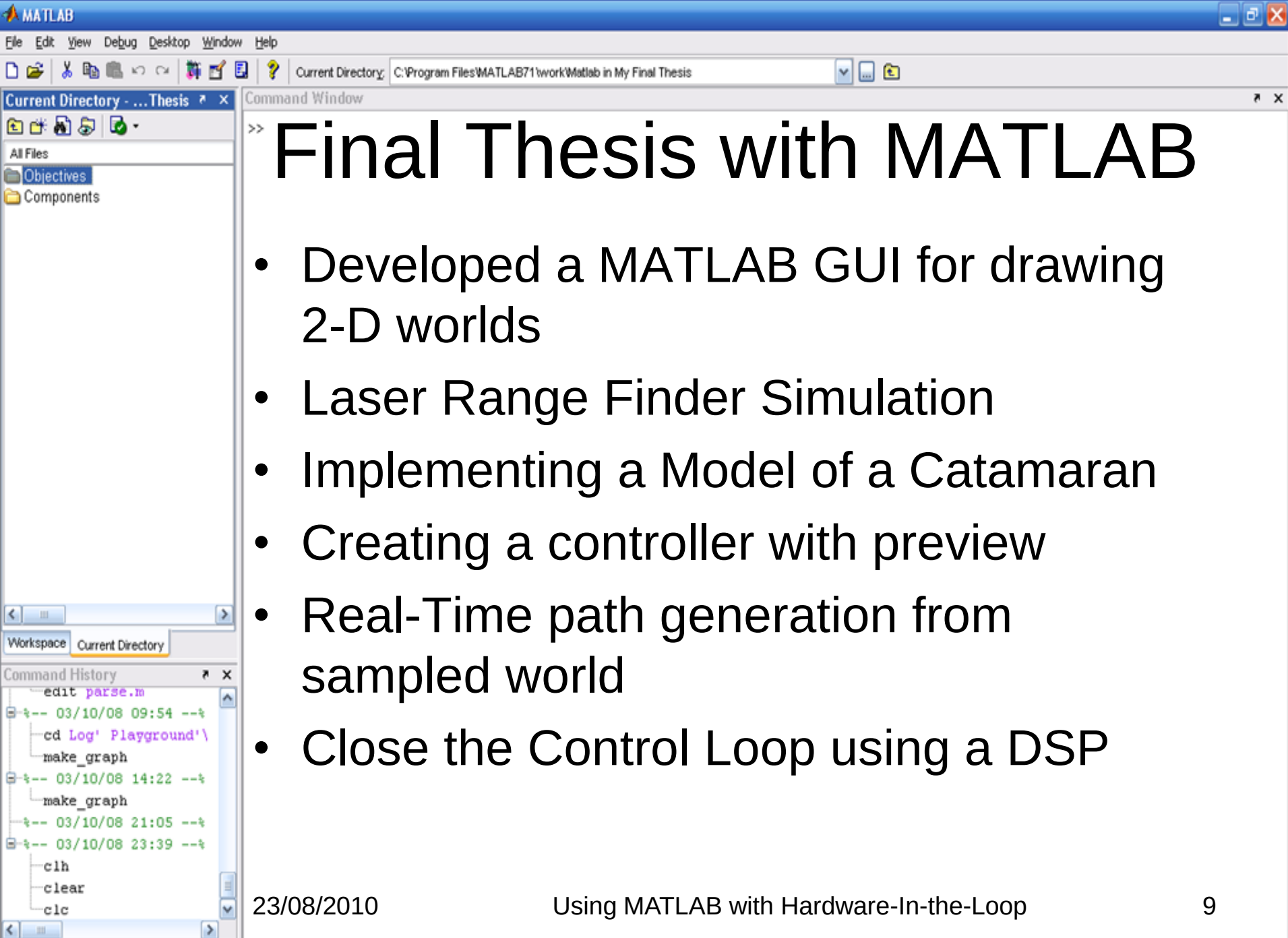
>> Final Thesis with MATLAB

- AUV running missions along a Coast
- Continuing an existing project
- Implementing a controller to control Catamaran
- Simulate how a Laser would help
- Learning about DSPs



23/08/2010

Using MATLAB with Hardware-In-the-Loop

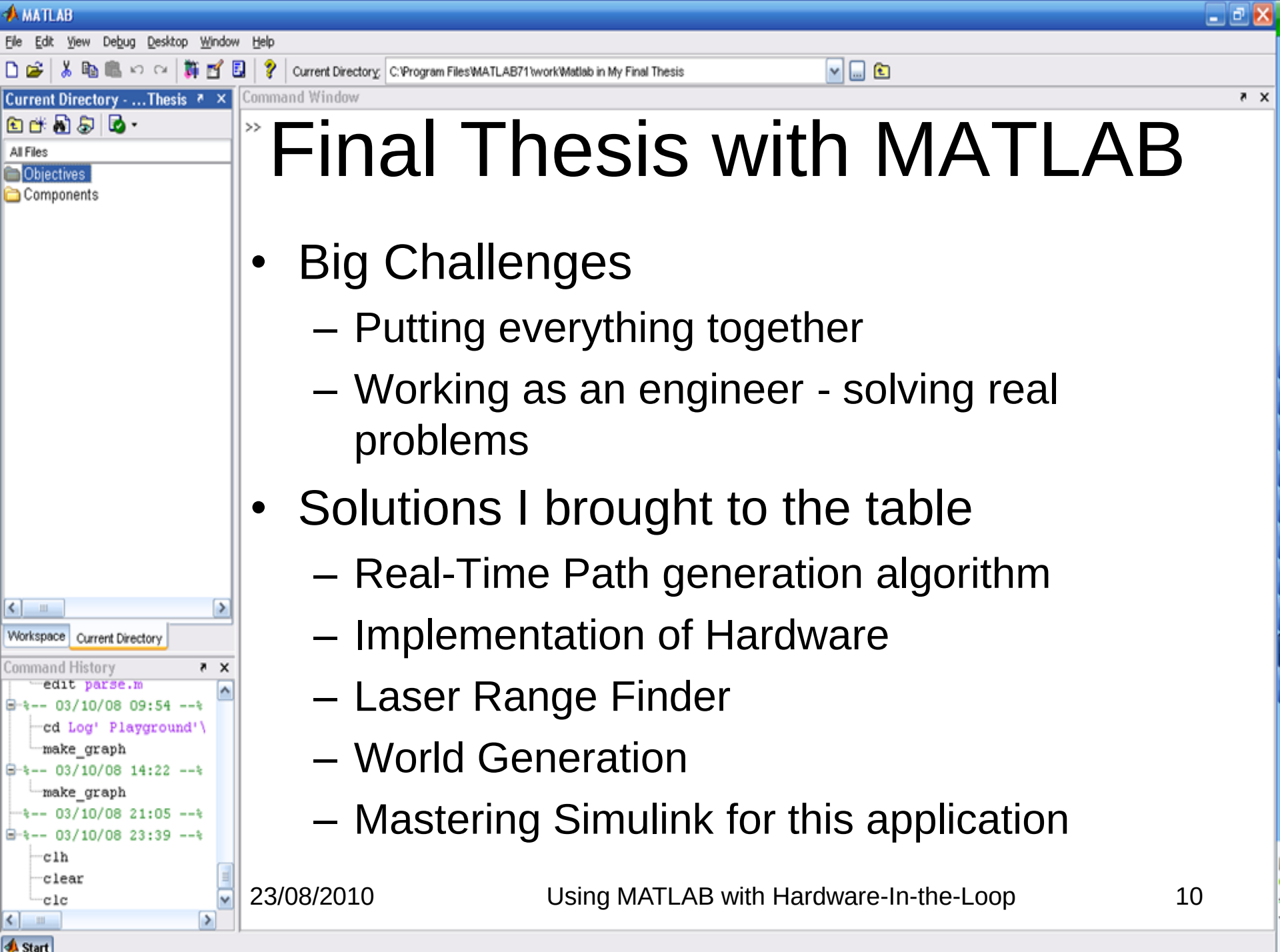


The image is a screenshot of the MATLAB software interface. The top menu bar includes File, Edit, View, Debug, Desktop, Window, and Help. The current directory is set to C:\Program Files\MATLAB71\work\Matlab in My Final Thesis. The Command Window displays a presentation titled "Final Thesis with MATLAB". The presentation lists six bullet points: "Developed a MATLAB GUI for drawing 2-D worlds", "Laser Range Finder Simulation", "Implementing a Model of a Catamaran", "Creating a controller with preview", "Real-Time path generation from sampled world", and "Close the Control Loop using a DSP". The left sidebar shows the file explorer with "Objectives" and "Components" folders. The Command History window at the bottom left shows a list of commands executed, including "edit parse.m", "cd Log\ Playground\'", "make_graph", "clear", and "clc".

Final Thesis with MATLAB

- Developed a MATLAB GUI for drawing 2-D worlds
- Laser Range Finder Simulation
- Implementing a Model of a Catamaran
- Creating a controller with preview
- Real-Time path generation from sampled world
- Close the Control Loop using a DSP

23/08/2010 Using MATLAB with Hardware-In-the-Loop

The image shows a MATLAB software window. On the left, there is a file explorer pane titled 'Current Directory - ...Thesis' showing a tree view with 'All Files', 'Objectives', and 'Components'. Below it is a 'Command History' pane showing a list of commands: 'edit parse.m', 'cd Log\ Playground\'', 'make_graph', 'make_graph', 'clear', and 'clc'. The main area of the window is a 'Command Window' which contains a large presentation slide. The slide has a title 'Final Thesis with MATLAB' and a bulleted list of challenges and solutions. The date '23/08/2010' and the text 'Using MATLAB with Hardware-In-the-Loop' are at the bottom of the slide. The MATLAB window's title bar and menu bar are visible at the top.

Final Thesis with MATLAB

- Big Challenges
 - Putting everything together
 - Working as an engineer - solving real problems
- Solutions I brought to the table
 - Real-Time Path generation algorithm
 - Implementation of Hardware
 - Laser Range Finder
 - World Generation
 - Mastering Simulink for this application

23/08/2010 Using MATLAB with Hardware-In-the-Loop

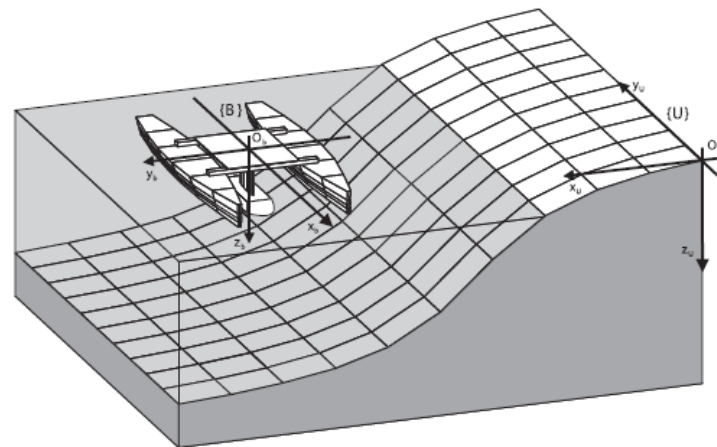
```

edit parse.m
-- 03/10/08 09:54 --
cd Log\ Playground\
make_graph
-- 03/10/08 14:22 --
make_graph
-- 03/10/08 21:05 --
-- 03/10/08 23:39 --
clh
clear
clc

```

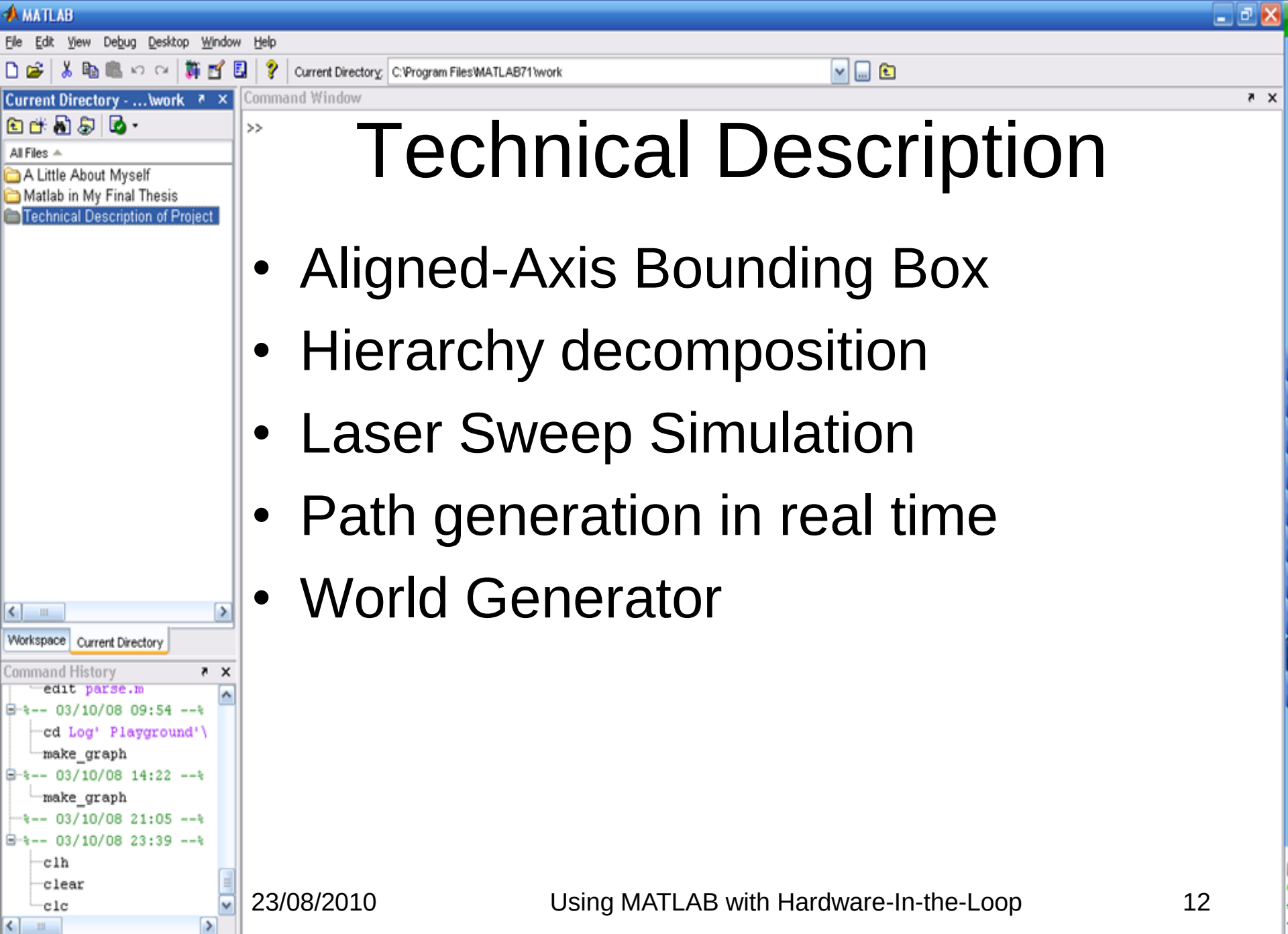
>> Final Thesis with MATLAB

- Catamaran Model
- Error Space
- Preview
- Control
- Developed Algorithms
- Implementation
- Results



$$\mathbf{x}_e = \begin{bmatrix} \mathbf{v}_e \\ r_e \\ d_t \\ \psi_e \end{bmatrix} = \begin{bmatrix} \mathbf{v}_b - {}^b_t R \mathbf{v}_r \\ r - (r_r + {}^t_r r_c) \\ [0 \ 1] {}^t_u R ({}^u \mathbf{p}_b - {}^u \mathbf{p}_c) \\ \psi_b - \psi_c \end{bmatrix}$$

DLQR



The image shows a MATLAB software interface. The top menu bar includes File, Edit, View, Debug, Desktop, Window, and Help. Below the menu bar is a toolbar with various icons. The current directory is set to C:\Program Files\MATLAB71\work. The Command Window displays the title 'Technical Description' and a list of five bullet points. The left sidebar shows a file explorer with a tree view containing folders like 'A Little About Myself', 'Matlab in My Final Thesis', and 'Technical Description of Project'. The bottom panel shows the Command History with a list of commands and timestamps.

File Edit View Debug Desktop Window Help

Current Directory: C:\Program Files\MATLAB71\work

Current Directory - ...work

Command Window

Technical Description

- Aligned-Axis Bounding Box
- Hierarchy decomposition
- Laser Sweep Simulation
- Path generation in real time
- World Generator

Workspace Current Directory

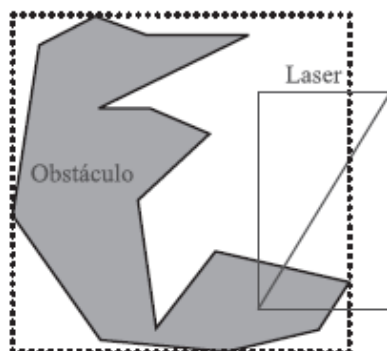
Command History

```
edit parse.m
-- 03/10/08 09:54 --
cd Log\ Playground\
make_graph
-- 03/10/08 14:22 --
make_graph
-- 03/10/08 21:05 --
-- 03/10/08 23:39 --
clh
clear
clc
```

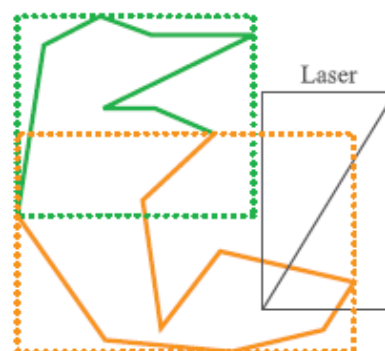
23/08/2010 Using MATLAB with Hardware-In-the-Loop

>>

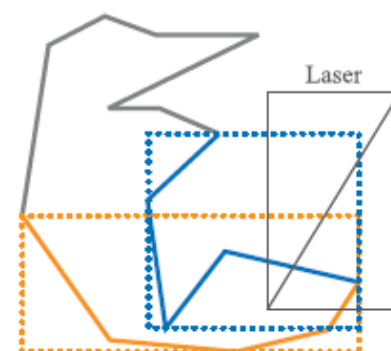
Developed Algorithms



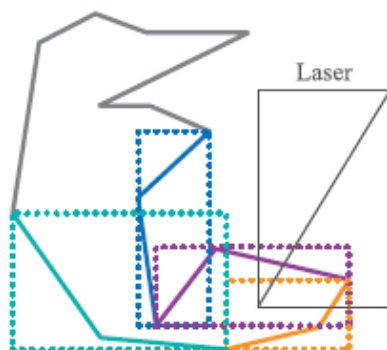
1



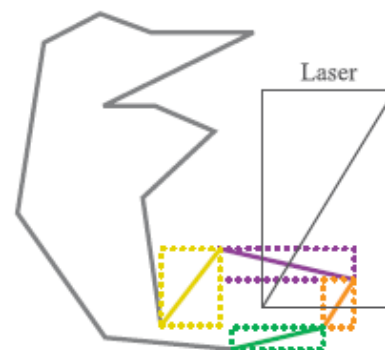
2



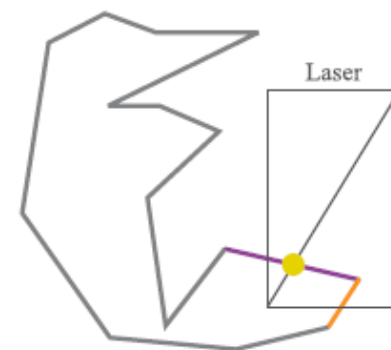
3



4



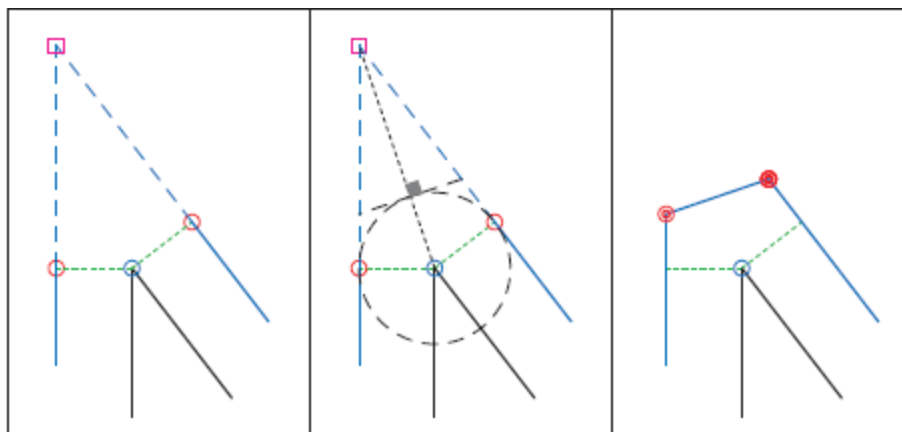
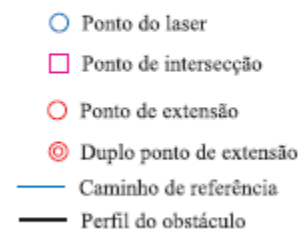
5

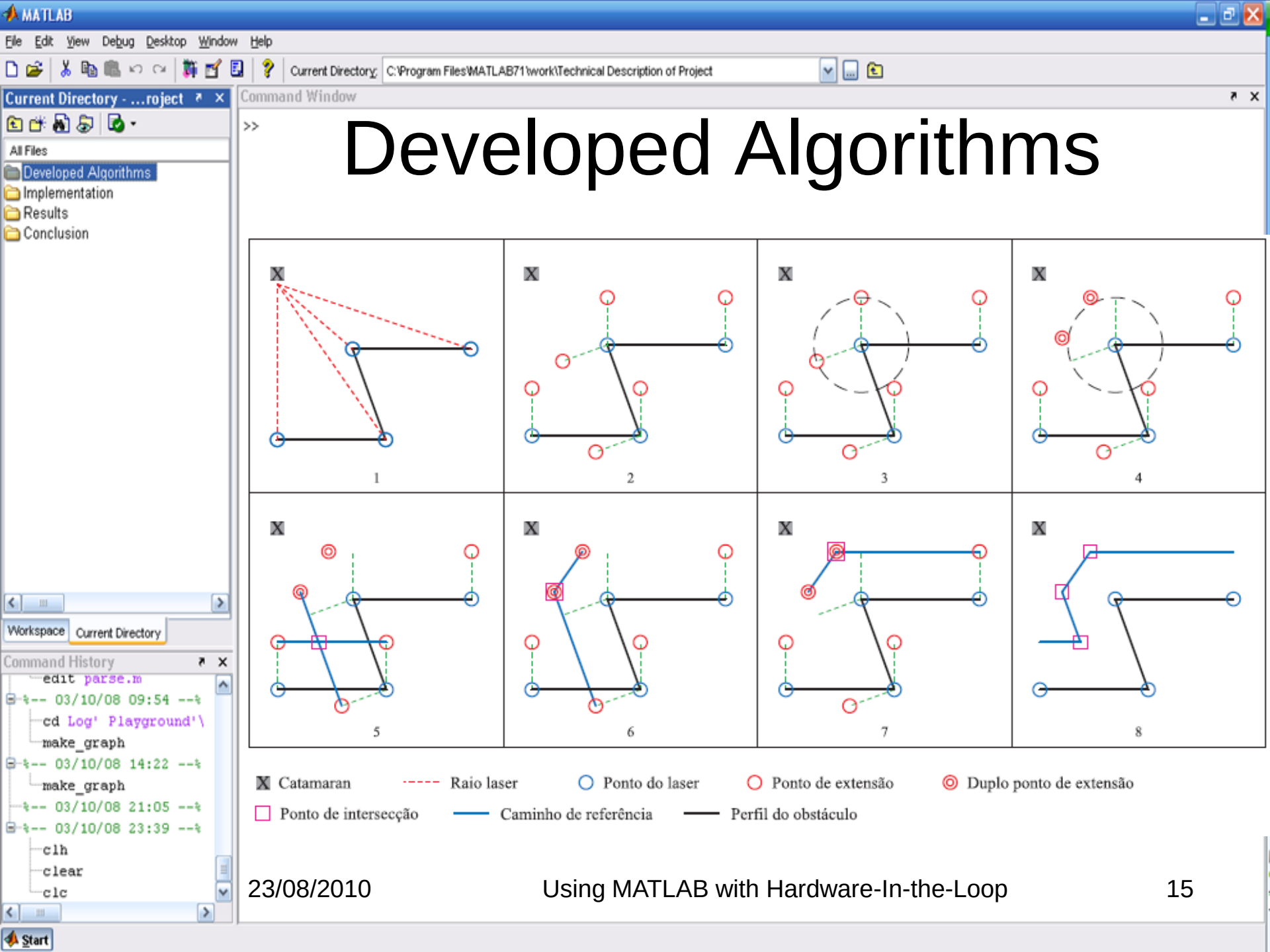


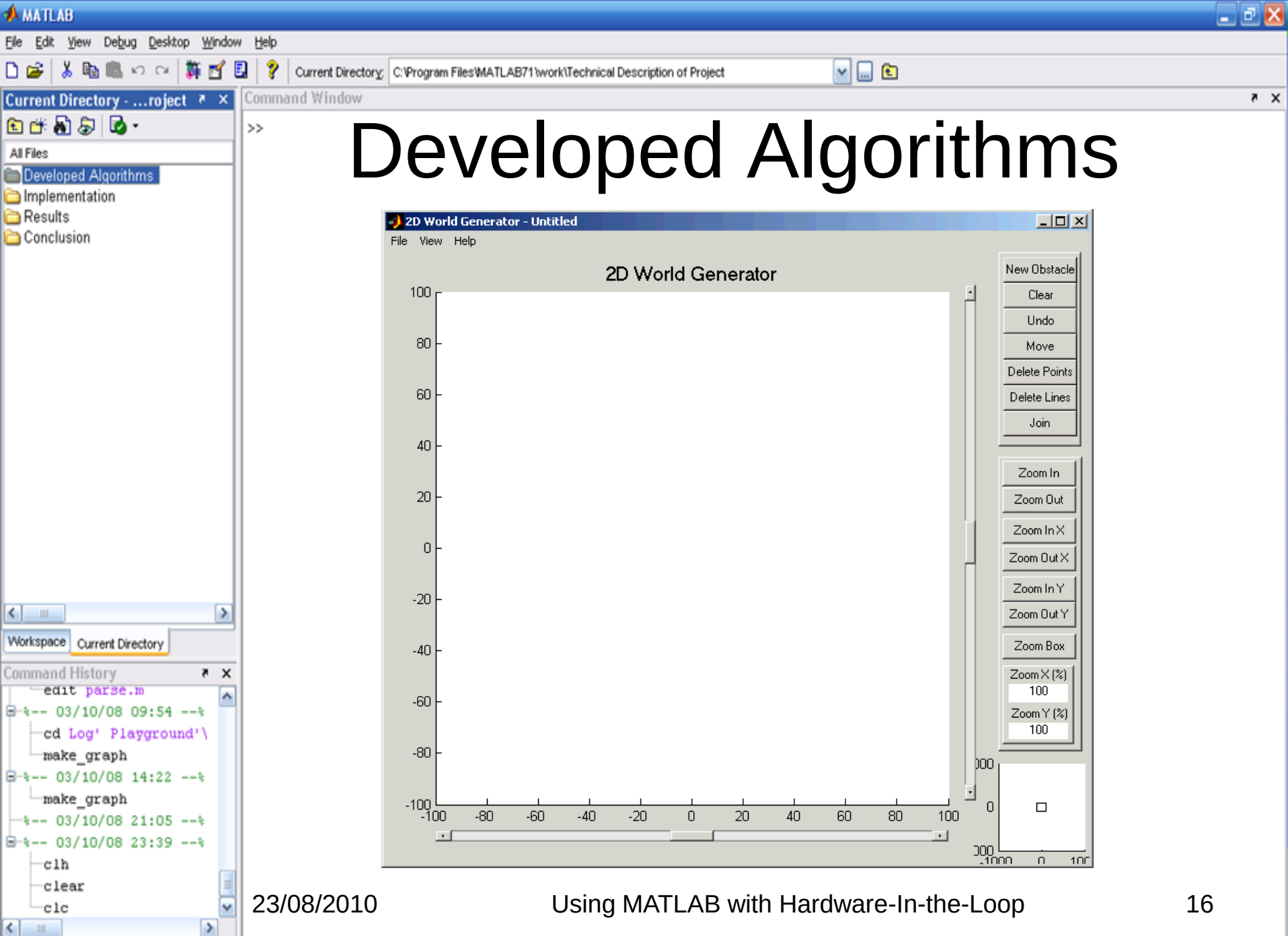
6

Command Window

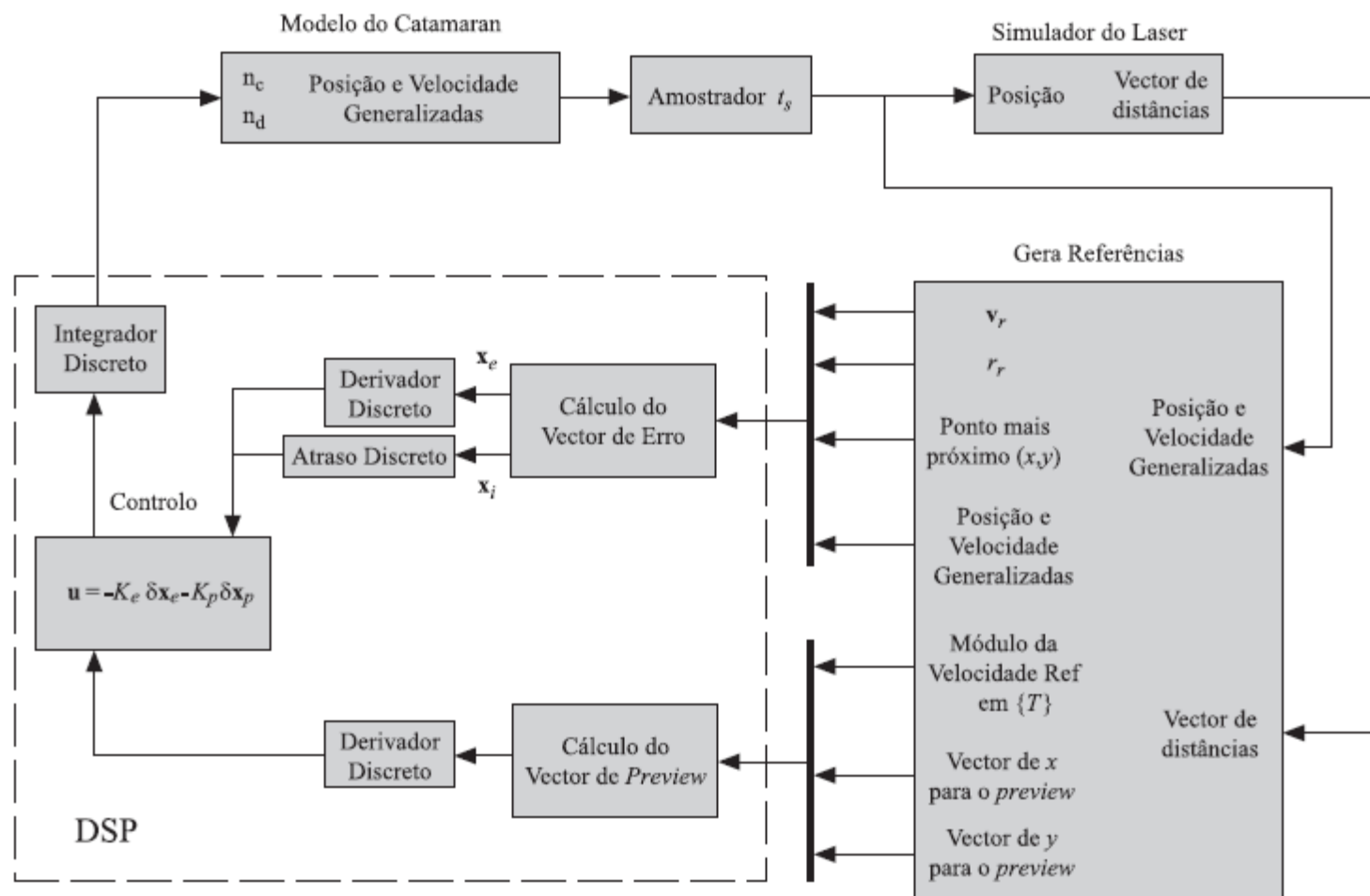
>>





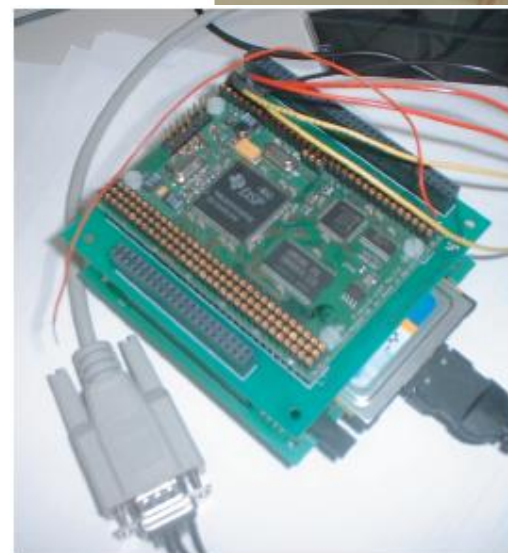
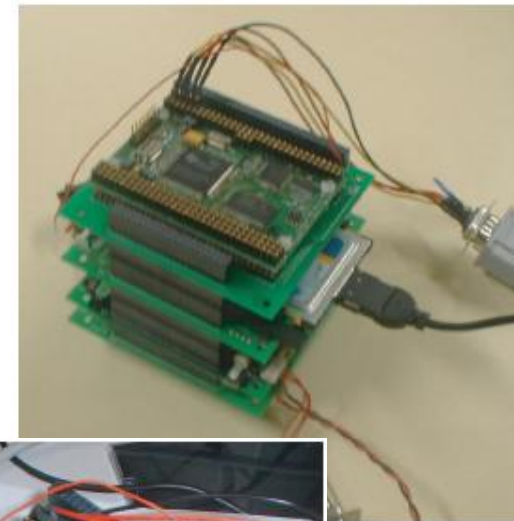


Implementation



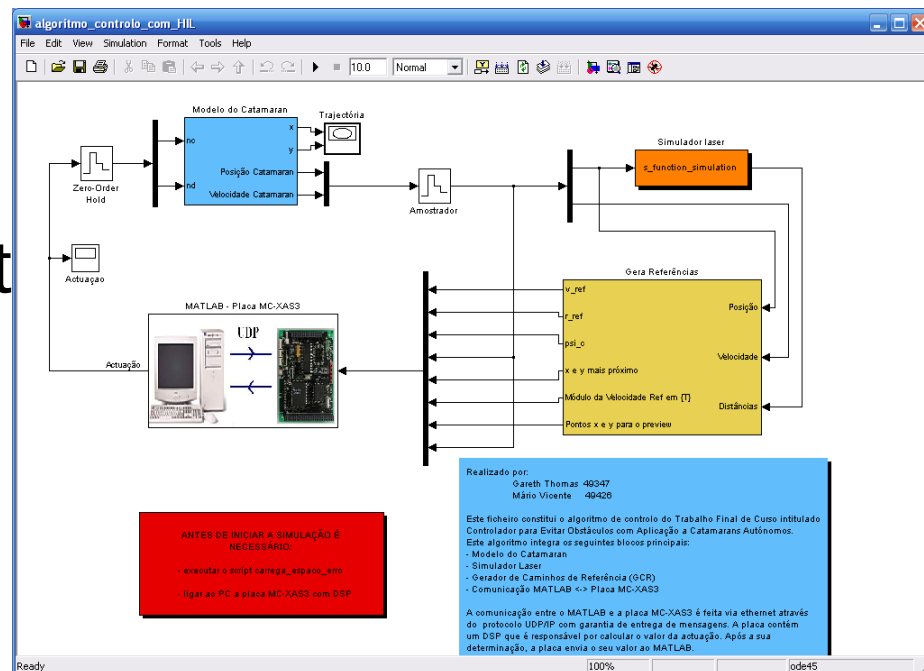
Implementation

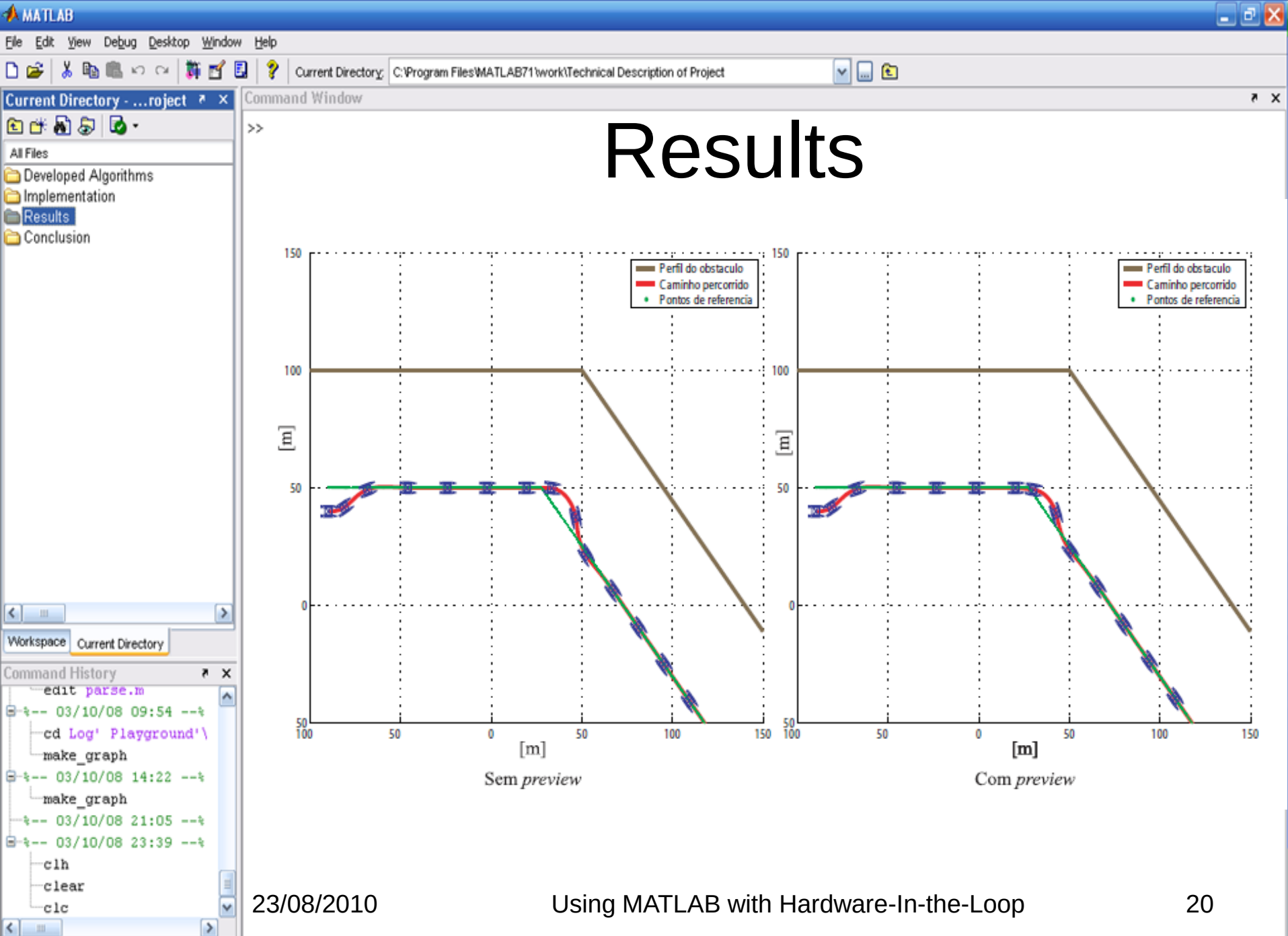
- PC with MATLAB
- MC-XAS3
- DSP TMS320VC33 – 32 bit
- DSP Interface
- PCMCIA
- DC Power Supply
- Hub

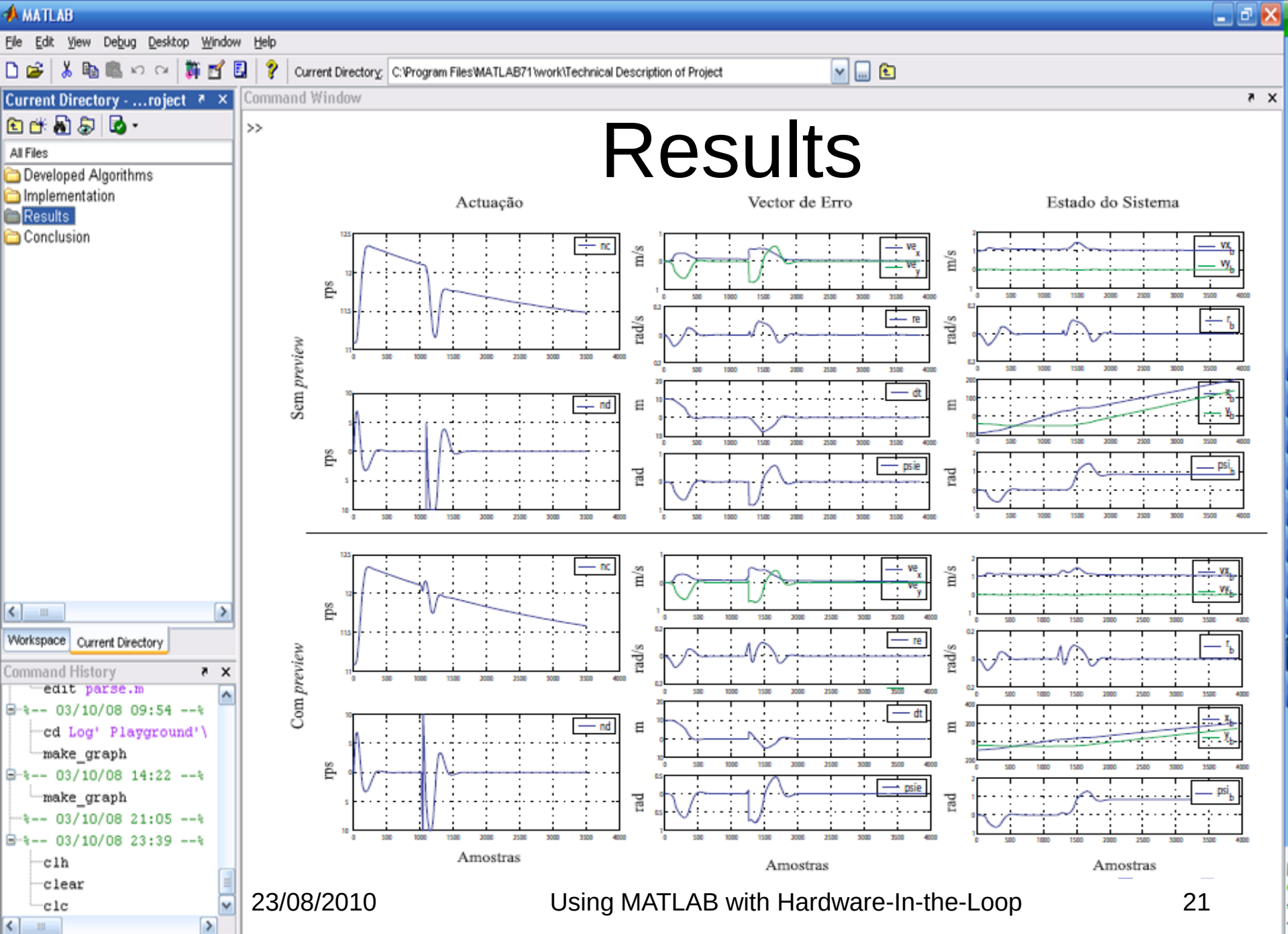


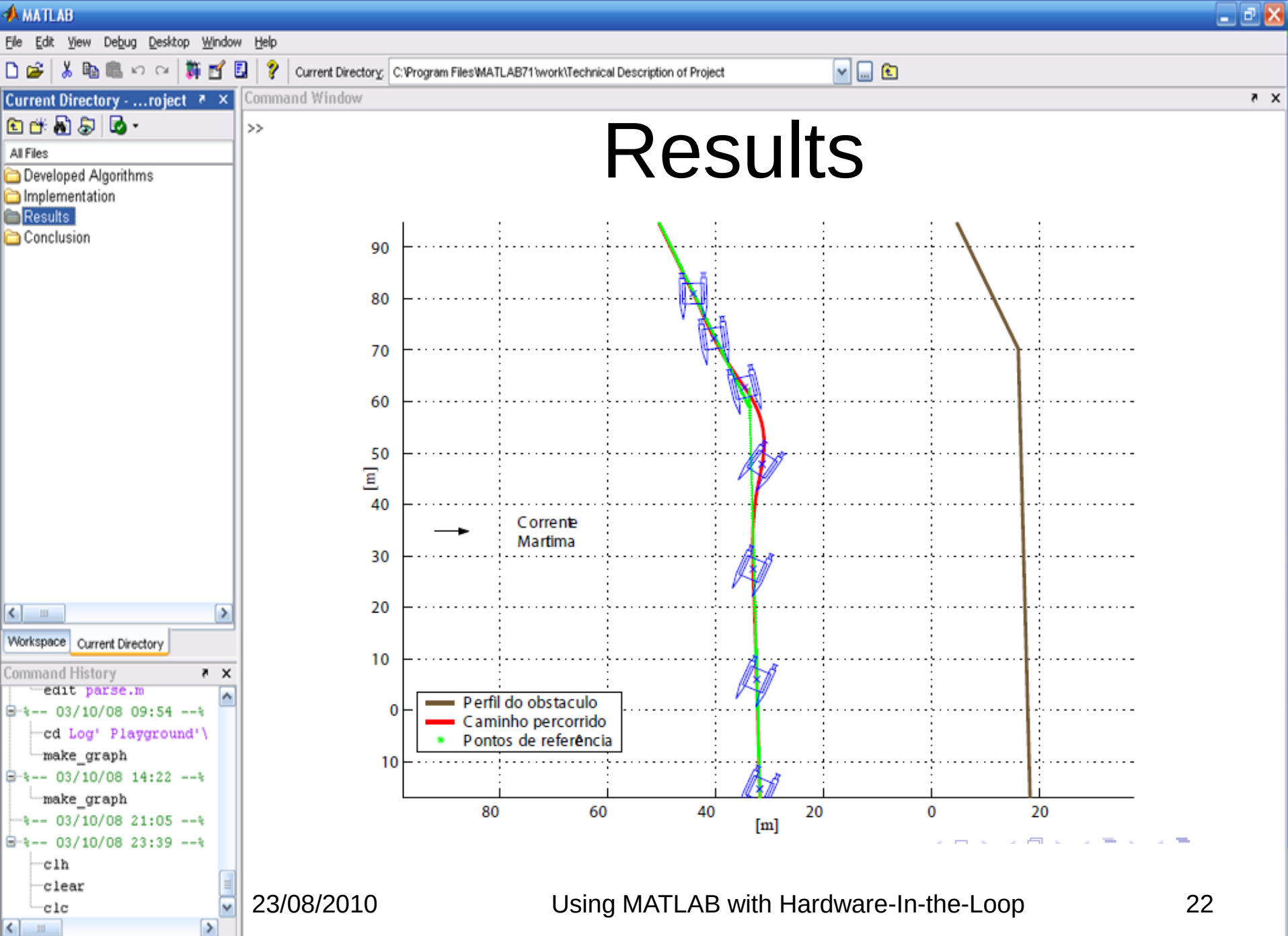
Implementation

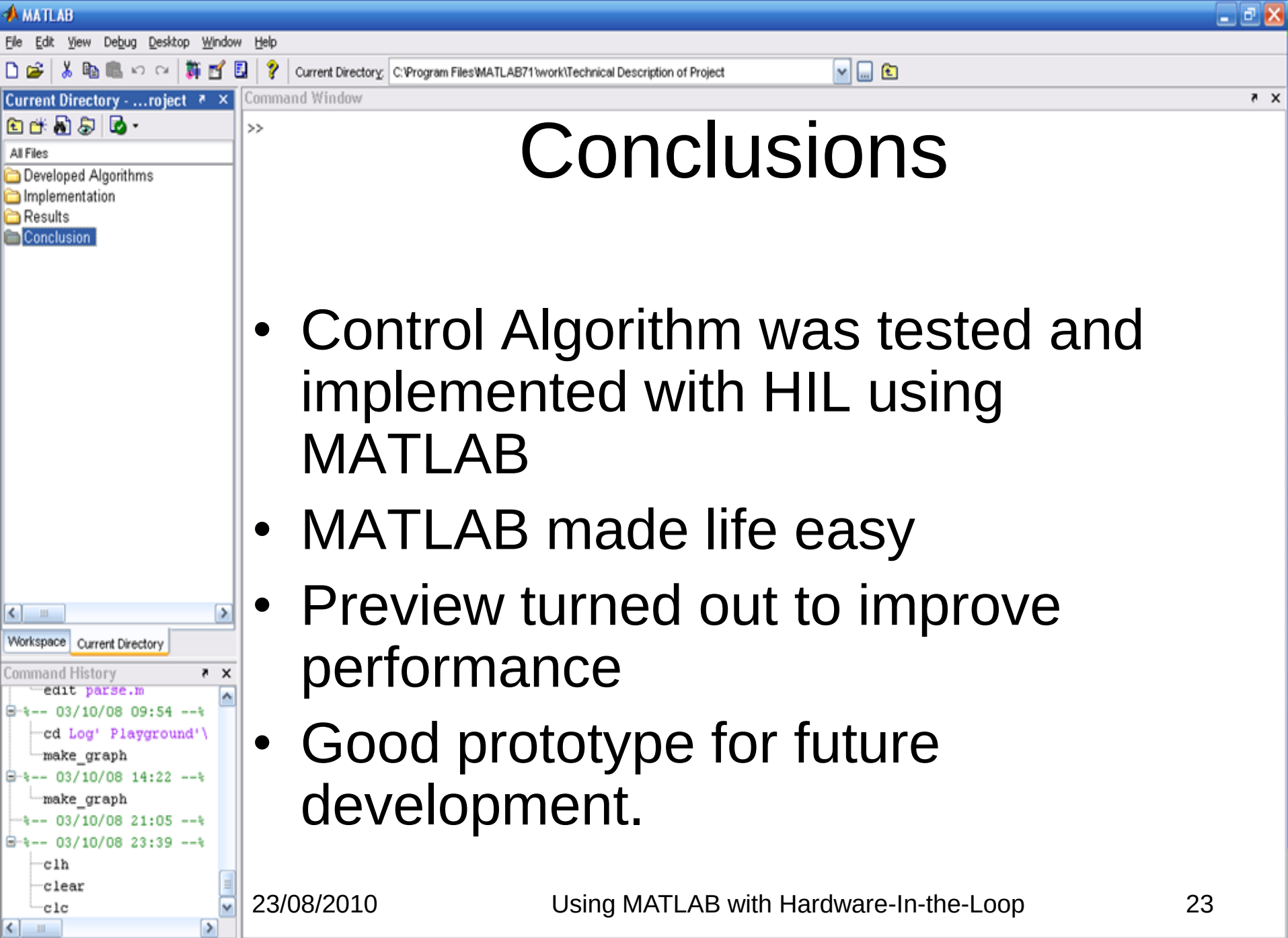
- Matlab to MCXAS3
 - UDP packet protocol
- MCXAS3 to DSP
 - Mailbox communication though shared memory









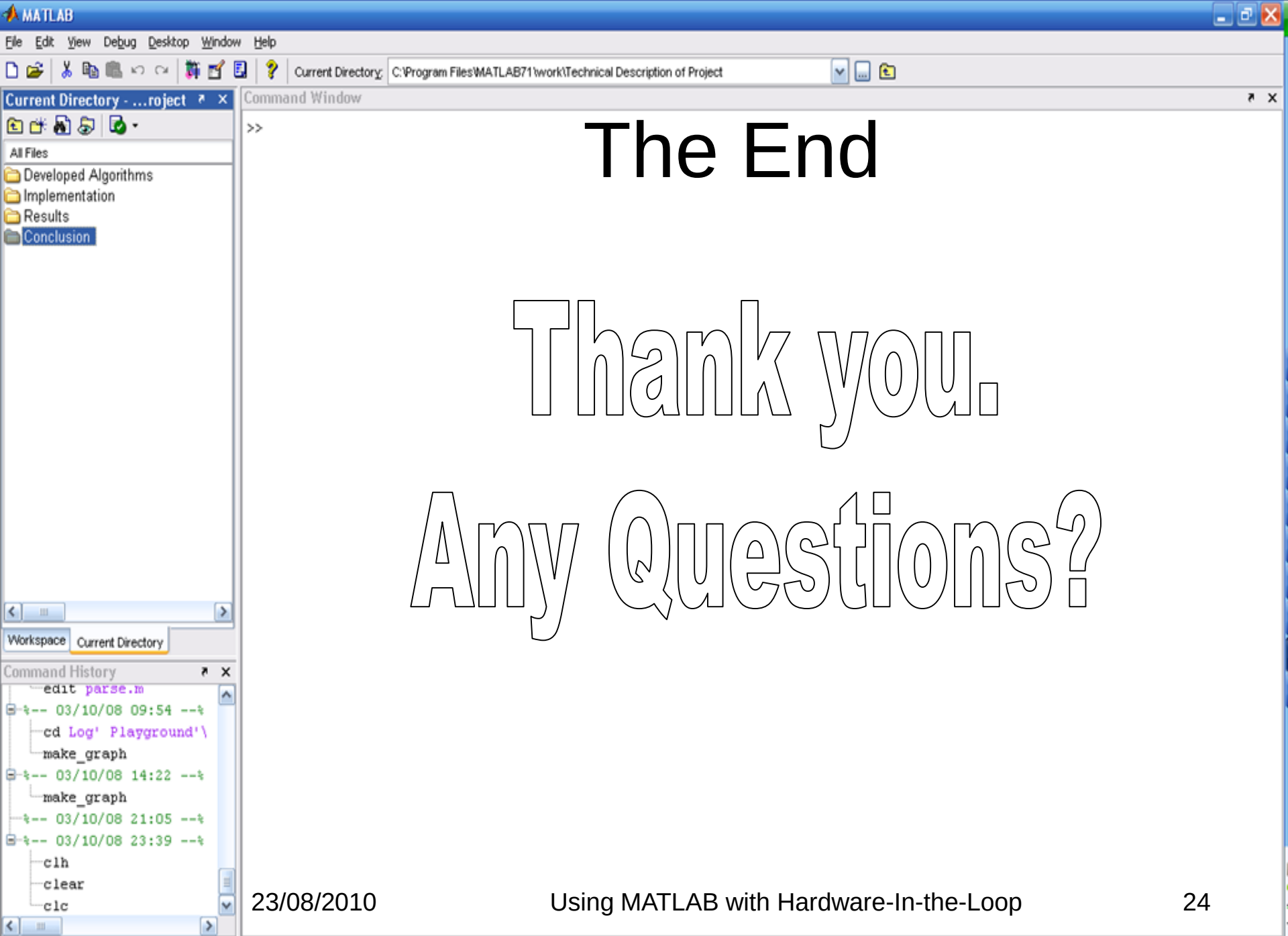


The image shows a MATLAB software interface. On the left, the 'Current Directory' pane shows a folder structure with 'Conclusion' selected. Below it, the 'Command History' pane shows a list of commands including 'edit parse.m', 'cd Log\ Playground\'', 'make_graph', 'clear', and 'clc'. The main area on the right is a 'Command Window' displaying a presentation slide. The slide has a title 'Conclusions' and a bulleted list of four points. The MATLAB window title is 'MATLAB' and the menu bar includes 'File', 'Edit', 'View', 'Debug', 'Desktop', 'Window', and 'Help'. The current directory path is 'C:\Program Files\MATLAB71\work\Technical Description of Project'.

Conclusions

- Control Algorithm was tested and implemented with HIL using MATLAB
- MATLAB made life easy
- Preview turned out to improve performance
- Good prototype for future development.

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The End

Thank you.
Any Questions?

23/08/2010

Using MATLAB with Hardware-In-the-Loop

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